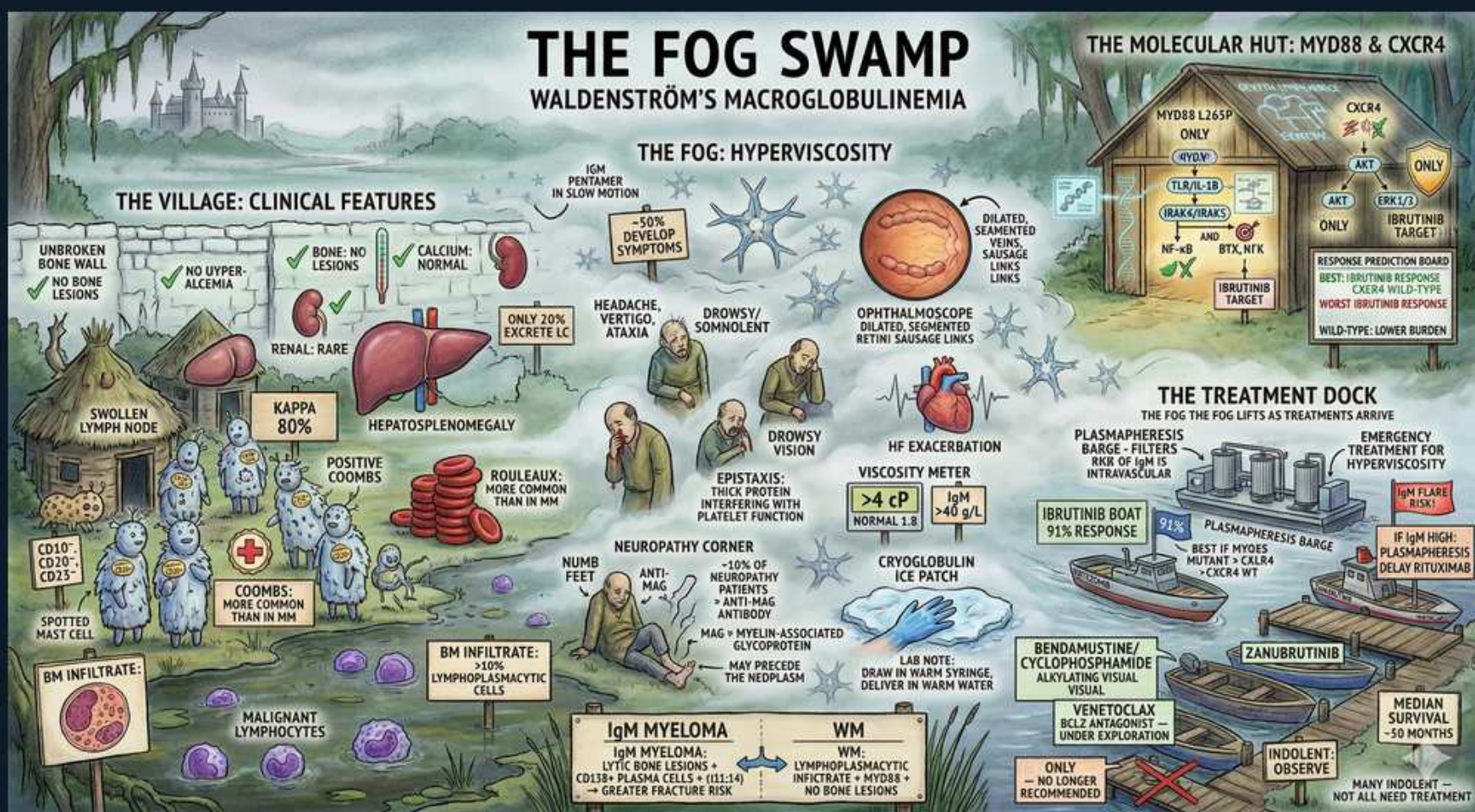


Waldenström's Macroglobulinemia — The Fog Swamp



1. IgM monoclonal protein

WHAT YOU SEE

Fog rolling over the swamp village.

WHAT IT ENCODES

WM = lymphoplasmacytic lymphoma secreting monoclonal IgM. Large pentameric IgM stays intravascular and drives hyperviscosity. Defined by IgM M-spike + bone-marrow lymphoplasmacytic infiltration.

BOARD TRAP

Don't call it multiple myeloma. MM secretes IgG/IgA/light-chain and causes lytic bone lesions; WM secretes IgM with NO lytic lesions and NO hypercalcemia.

2. Hyperviscosity syndrome

WHAT YOU SEE

Thick fog, slow-moving figures.

WHAT IT ENCODES

High serum IgM raises viscosity: headache, dizziness, ataxia, drowsiness, blurred vision. Symptoms typically appear once serum viscosity >4 centipoise. The classic mucosal bleeding plus neuro triad.

BOARD TRAP

Hyperviscosity is the WM emergency, not tumor bulk. Symptoms correlate with IgM/viscosity level, not marrow plasma-cell percentage.

3. Retinal 'sausage' veins

WHAT YOU SEE

Ophthalmoscope showing dilated, segmented retinal veins.

WHAT IT ENCODES

Fundoscopy shows dilated, tortuous, segmented ('sausage-link' / boxcar) retinal veins, flame hemorrhages, papilledema — the hallmark sign of hyperviscosity.

BOARD TRAP

Sausage-link retinal veins point to IgM hyperviscosity (WM), not diabetic or hypertensive retinopathy.

4. MYD88 L265P mutation

WHAT YOU SEE

Molecular hut labeled MYD88.

WHAT IT ENCODES

MYD88 L265P is present in >90% of WM and activates NF- κ B via BTK signaling. It is the rationale for BTK-inhibitor therapy (ibrutinib) and helps distinguish WM from other IgM gammopathies.

BOARD TRAP

MYD88 mutant + CXCR4 WT predicts the best ibrutinib response; CXCR4 mutation blunts and delays ibrutinib response.

5. CXCR4 mutation

WHAT YOU SEE

Side room of the hut labeled CXCR4.

WHAT IT ENCODES

CXCR4 WHIM-like mutations occur in ~30-40% of WM. They confer drug resistance and predict delayed/blunted response to ibrutinib, and higher IgM/hyperviscosity burden.

BOARD TRAP

CXCR4 mutation = worse, slower ibrutinib response — the opposite of MYD88-mutant/CXCR4-WT patients.

6. Ibrutinib (BTK inhibitor)

WHAT YOU SEE

Ibrutinib boat at the treatment dock.

WHAT IT ENCODES

Ibrutinib blocks BTK downstream of mutant MYD88; ~91% response in WM. First-line oral option, especially for symptomatic disease or hyperviscosity not needing immediate plasmapheresis.

BOARD TRAP

Response depends on genotype: best in MYD88-mut/CXCR4-WT, weakest in MYD88-WT.

7. Plasmapheresis

WHAT YOU SEE

Barge filtering the fog from the swamp.

WHAT IT ENCODES

Plasmapheresis rapidly removes intravascular IgM and is FIRST-LINE for acute symptomatic hyperviscosity. It is a bridge — it does not treat the clone, so definitive therapy must follow.

BOARD TRAP

For acute hyperviscosity, plasmapheresis comes BEFORE/with starting chemo — don't wait for rituximab to work.

8. Avoid single-agent rituximab (IgM flare)

WHAT YOU SEE

Lab note: rituximab may cause IgM flare.

WHAT IT ENCODES

Rituximab can trigger a transient IgM 'flare' (rise) that worsens hyperviscosity. In patients with high IgM, hold rituximab or plasmapheresis first / use combination regimens.

BOARD TRAP

Giving rituximab alone to a high-IgM WM patient can precipitate hyperviscosity crisis — a classic exam pitfall.

9. Bendamustine–rituximab / DRC

WHAT YOU SEE

Visual aid + alkylating treatment crate.

WHAT IT ENCODES

Chemoimmunotherapy (bendamustine-rituximab, or dexamethasone-rituximab-cyclophosphamide) is standard for symptomatic WM, especially with bulky disease. Alkylator-based regimens give durable responses.

BOARD TRAP

Asymptomatic WM is observed; treat only for symptoms (hyperviscosity, anemia, B-symptoms), not for IgM level alone.

10. Cold agglutinins / cryoglobulins

WHAT YOU SEE

Cryoglobulin ice patch, cold-reactive antibody.

WHAT IT ENCODES

Monoclonal IgM can act as a cold agglutinin (Coombs positive, cold-reactive) causing hemolysis, or as type I cryoglobulin causing acrocyanosis. Warm the patient/keep samples warm.

BOARD TRAP

WM hemolysis is cold-agglutinin mediated (IgM, C3+ DAT, worse in cold) — not warm AIHA.

11. Anemia (most common cytopenia)

WHAT YOU SEE

Pale, tired villagers; positive Coombs.

WHAT IT ENCODES

Anemia is the most common indication to treat WM, from marrow infiltration plus dilutional effect of high IgM expanding plasma volume. Often with fatigue and constitutional symptoms.

BOARD TRAP

Anemia in WM is multifactorial (marrow + dilution from plasma volume expansion), so Hb can look falsely low.

12. Hepatosplenomegaly / lymphadenopathy

WHAT YOU SEE

Enlarged spleen and nodes in the village.

WHAT IT ENCODES

Because WM is a lymphoplasmacytic LYMPHOMA, organomegaly and lymphadenopathy occur — features myeloma lacks. Reflects tissue infiltration by the lymphoplasmacytic clone.

BOARD TRAP

Lymphadenopathy/HSM favor WM over myeloma; their presence argues against pure plasma-cell myeloma.

13. Peripheral neuropathy (anti-MAG)

WHAT YOU SEE

Numb feet corner, anti-MAG antibody.

WHAT IT ENCODES

IgM can target myelin-associated glycoprotein (anti-MAG), producing a distal demyelinating sensory neuropathy. Distinct mechanism from amyloid or compressive neuropathy.

BOARD TRAP

WM neuropathy is often anti-MAG demyelinating — test for it; it differs from POEMS or amyloid neuropathy.

14. BM lymphoplasmacytic infiltrate

WHAT YOU SEE

Marrow slide with lymphoplasmacytic cells.

WHAT IT ENCODES

Diagnosis requires ≥10% clonal lymphoplasmacytic cells in marrow plus IgM monoclonal protein of any level. Cells show CD19/CD20+ with plasmacytoid differentiation.

BOARD TRAP

WM marrow is lymphoplasmacytic (small B-cells + plasma cells), not sheets of pure plasma cells as in myeloma.

15. No lytic bone lesions

WHAT YOU SEE

Sign: 'No bone lesions, normal calcium.'

WHAT IT ENCODES

Unlike myeloma, WM characteristically spares bone — no lytic lesions, no hypercalcemia, and renal involvement is rare. A key MM-vs-WM discriminator.

BOARD TRAP

If the IgM gammopathy patient has lytic lesions/hypercalcemia, reconsider IgM myeloma (rare) rather than WM.